

EFFICIENT AND SCALABLE TOBIT GAUSSIAN PROCESS REGRESSION FOR MODELING AIR QUALITY DATA

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ABSTRACT

Tobit Gaussian process regression (T-GPR) model extends the classical Tobit regression model by incorporating the flexibility of Gaussian processes, thereby enabling the characterization of complex, nonlinear response-covariate relationships under censoring. However, the naive implementations of T-GPR is constrained by cubic computational cost and quadratic memory requirements, limiting their applicability in large-scale environmental applications. In this article, we develop a fast T-GPR method based on a sparse ordered conditional approximation of latent Gaussian processes to achieve linear time and space complexity. Furthermore, we present conditions guaranteeing posterior contraction within the proposed framework. The practical utility of the proposed methodology is showcased via numerical experiments and analysis of nitrogen oxide concentration data from the London Air Quality Archive.

Index Terms— Gaussian process regression, Posterior contraction, Tobit modeling, NO_x concentration data, Vecchia approximation.

1. INTRODUCTION

Tobit Gaussian process regression (T-GPR) [1, 2] is well-suited for modeling censored observations due to its ability to capture nonlinear relationships between responses and covariates. Given a functional regression model $y = f(x) + \varepsilon$, T-GPR extends the classical Tobit framework [3] by introducing an auxiliary variable z such that $z \geq \tau$ whenever y is censored (right: $y \geq \tau$, left: $y \leq \tau$). The latent function f is modeled as a Gaussian process, $f(\cdot) \sim \mathcal{N}(0, \sigma_0^2 k_\theta(\cdot, \cdot))$, where $k_\theta : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a positive definite kernel with unit variance controlling the shape of the function. Despite its flexibility and wide applicability, T-GPR inherits the cubic computational cost of Gaussian processes, limiting its use for large datasets.

Several approaches have been proposed to mitigate the cubic computational cost of Gaussian processes, in-

cluding basis function approximations, covariance tapering [4], sparse Cholesky factorizations [5, 6], sparse variational methods [7], and inducing point techniques [8]. Basis function based methods reduce complexity by low-rank approximations, while sparse Cholesky factorizations exploit sparsity in covariance or precision matrices to achieve $\mathcal{O}(nr)$ memory and $\mathcal{O}(nr^2)$ time for $r \ll n$. The Vecchia approximation [9] and its extensions [5, 10, 6] factorize the joint distribution into sparse ordered conditionals that can be computed in parallel. This results in sparse Cholesky factorizations with linear memory scaling, making the method highly scalable. Consequently, Vecchia-based approaches have been adopted in plethora of applications [11].

To the best of our knowledge, [12] and [13] are the only works addressing scalable T-GPR. [12] employs a variational posterior, and its computation scales cubically with the number of inducing points, making it impractical when a large number of inducing points is required [7]. [13] uses a nearest neighbor Gaussian process (NNGP)-based Tobit regression model for censored functional data analysis. Moreover, neither approach provides theoretical guarantees, motivating the development of a computationally efficient and theoretically grounded T-GPR framework.

In this article, we propose Vecchia-assisted GPR (VaT-GPR), a scalable extension of T-GPR for censored functional responses that leverages sparse Cholesky factorization and parallel computation to achieve linear memory and computational cost. We establish conditions for posterior contraction and demonstrate the utility of the method via simulations and a real-data application.

2. PROPOSED METHODOLOGY

2.1. Tobit Gaussian Process Regression (T-GPR)

We denote the observed response and covariates by $\mathcal{D} = \{(y_i, x_i)\}_{i=1}^n$. In many applications, the observed response is subject to left censoring at a known threshold τ . Consequently, instead of observing the true latent value, denoted by z_i , we only observe $y_i = \max(z_i, \tau)$

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where $z_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma_0^2 g)$ and f is an unknown function capturing the underlying relationship between inputs x_i and latent variable z_i . f is often modeled via a Gaussian process (GP) $f \sim \mathcal{GP}(0, \sigma_0^2 k_l(\cdot, \cdot))$, where k_l is a Matérn covariance kernel with range l and variance set to 1 to ensure identifiability in the marginal distribution of z_i . Because observations are censored from below at τ , the likelihood contribution of each data point depends on whether it is censored or not:

$$p(y_i | f(x_i), \sigma_0^2) = \begin{cases} \mathcal{N}(y_i; f(x_i), \sigma_0^2 g), & \text{if } y_i > \tau, \\ \Phi\left(\frac{\tau - f(x_i)}{\sigma_0 g}\right), & \text{if } y_i = \tau, \end{cases}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. Without loss of generality, we assume $y_i = \tau$ for $i = 1, \dots, p$. We further note that latent variables

$$z_i | f(x_i) \sim \delta_{y_i} \mathbf{1}(y_i > \tau) + \mathcal{TN}(\tau, \infty; f(x_i), \sigma_0^2 g) \mathbf{1}(y_i = \tau),$$

where δ_{y_i} is a point mass at y_i , and $\mathcal{TN}(\tau, \infty; \mu, \sigma_0^2 g)$ denotes a truncated normal distribution with mean μ , variance $\sigma_0^2 g$, truncated below at τ . We aim to perform Bayesian inference on $(p + 3)$ quantities $\boldsymbol{\theta} = (z_{1:p}, l, \sigma_0^2, g)$ using a Gibbs sampling algorithm.

The key computational bottleneck involves the matrix inversion required to evaluate the probability density function of $\mathcal{N}_n(\mathbf{0}, \sigma_0^2 \mathbf{K}_l)$, which incurs cubic time complexity and squared storage complexity. To elucidate the discussion, we assume $\mathbf{K}_l = \mathbf{K}$ for the time being. Motivated by the decomposition of the joint density of $\mathbf{f} = [f(x_1), \dots, f(x_n)]$ as $p(\mathbf{f}) = \prod_{i=1}^n p(f_i | \mathbf{f}_{1:i-1})$, [9] proposed $\hat{p}(\mathbf{f}) = \prod_{i=1}^n p(f_i | \mathbf{f}_{c(i)})$, where $c(i) \subset \{1, \dots, i-1\}$ is a conditioning index set of size $|c(i)| = \min(m, i-1) \forall i = 2, \dots, n$ and $c(1) = \emptyset$. This induces a sparse dependence graph, denoted by $\mathcal{S}$, of $\mathbf{f}$. Once the inputs $(x_1, \dots, x_n)$ are ordered using maxmin ordering [14], even a relatively small value of $m \ll n$ yields a highly accurate approximation $\hat{p}(\mathbf{f})$ of $p(\mathbf{f})$ due to the presence of screening effect [15]. Moreover, [16] showed that, under a sparsity pattern $\mathcal{S}$ on the precision matrix, the Vecchia approximation can be interpreted as the Kullback–Leibler projection of the full Gaussian distribution onto the set of Gaussian distributions with precision matrices respecting the given sparsity structure, i.e., $\hat{p}(\mathbf{f}) = \arg \min_{\mathbf{U} \in \mathbf{U}_{\mathcal{S}}} \text{KL}[\mathcal{N}_n(\mathbf{0}, \sigma_0^2 \mathbf{K}) \| \mathcal{N}_n(\mathbf{0}, (\mathbf{U}\mathbf{U}^\top)^{-1})]$, with $\mathbf{U}_{\mathcal{S}} = \{\mathbf{U} \in \mathcal{R}^{n \times n} : U_{ij} \neq 0 \implies (i, j) \in \mathcal{S}\}$. Further, the optimal Cholesky factor $\mathbf{U}^*$ can be computed in $\mathcal{O}(nm^3)$ time [16].

2.2. Vecchia-assisted Tobit GPR (VaT-GPR)

To propose our method, we replace the GP prior on $\mathbf{f}$ in Section 2.1 by its Vecchia approximation-based sparse inverse Cholesky decomposition. The details follows.

Likelihood and prior. We follow the maxmin ordering [14] to rearrange the points in the inputs, and fix the number of nearest neighbors, i.e., $|c(i)| \leq m \forall i = 1, \dots, n$ that introduces a sparsity pattern in the Cholesky factor of the precision matrices. For the rest of the paper, we use this sparsity to define a GP $\mathbf{f} \sim \mathcal{N}_n(\mathbf{0}, [\sigma_0^2 \mathbf{U}_l^* \mathbf{U}_l^{*\top}]^{-1})$, where $\mathbf{U}_l^*$ is the Vecchia-assisted sparse Cholesky factor of $\mathbf{K}_l^{-1}$. [16] and [6] compute $\mathbf{V}_l^* = \text{rchol}([\mathbf{U}_l^* \mathbf{U}_l^{*\top}] + g^{-1} \mathbf{I}_n)$, where $\text{rchol}(\mathbf{B}) = \mathbf{P} \text{chol}(\mathbf{P} \mathbf{B} \mathbf{P}) \mathbf{P}$, $\mathbf{P}$ denote a row-column reversing permutation matrix. We follow these computations to approximate $\mathbf{y} \sim \mathcal{N}_n(\mathbf{0}, \Sigma_{l, \mathcal{S}} = \sigma_0^2 g \mathbf{U}_l^* \mathbf{U}_l^{*\top} \mathbf{V}_l^* \mathbf{V}_l^{*\top})$.

As a starting point, we first fit a noisy Gaussian process $\mathcal{GP}(0, \sigma_0^2 (\mathbf{K}_l + g \mathbf{I}_n))$ on the uncensored responses $\mathcal{D}_{-p} = (x_i, y_i)_{i=p+1, \dots, n}$, and obtain initial estimates $\hat{\sigma}_0^2, \hat{l}, \hat{g}$. To conduct Bayesian inference, we assign relatively non-informative prior distributions for l and g such that we put non-zero prior mass to $\hat{l}$ and $\hat{g}$. Further, we consider $\log(\sigma_0^2)$, i.e., $\pi(\log \sigma_0^2) \equiv 1$. This completes the hierarchical model and prior specification.

Gibbs sampler. We next present a Gibbs sampler algorithm to draw random samples from the joint posterior distribution of $[f(\cdot), z_1, \dots, z_p, l, g, \sigma_0^2 | \mathbf{y}]$ through the following steps.

(a) Sampling from $[f(\cdot) | \mathbf{z}, l, g, \sigma_0^2]$. We follow [10] and [6] to compute the posterior distribution $f | \mathbf{z}, l, g, \sigma_0^2$ expressed as, $\mathbf{f} | \mathbf{z}, l, g, \sigma_0^2 \sim \mathcal{N}_n(\tilde{\boldsymbol{\mu}}, \tilde{\mathbf{K}})$, with $\tilde{\boldsymbol{\mu}} = \mathbf{V}_l^{*-1} \mathbf{y}$, $\tilde{\mathbf{K}} = [\mathbf{V}_l^* \mathbf{V}_l^{*\top}]^{-1}$.

(b) Sampling from $[l, g | \mathbf{z}, \sigma_0^2]$. We use the integrated likelihood $[\mathbf{z} | l, g, \sigma_0^2]$ to design Metropolis-Hastings sampling strategy for both l and g . For updating l and g , we consider a uniform proposal density, and employ a M–H acceptance/rejection scheme.

(c) Sampling from $[\sigma_0^2 | f(\cdot), \mathbf{z}, l, g]$. It is straightforward to show that $\sigma_0^2 | l, g, \mathbf{z} \sim \mathcal{IG}(n/2, \frac{\mathbf{z}^\top \Sigma_{l, \mathcal{S}}^{-1} \mathbf{z}}{2})$, where $\mathcal{IG}(a, b)$ denote an inverse Gamma distribution.

(d) Sampling from $[\mathbf{z} | f(\cdot), l, y, g, \sigma_0^2]$. We recall that we have only the first p responses as censored. Hence, we update $z_i, 1 \leq i \leq p$ as

$$z_i | f(x_i), \sigma_0^2, g \sim \mathcal{TN}(\tau, \infty; f(x_i), \sigma_0^2 g).$$

3. POSTERIOR CONTRACTION

A general framework for posterior contraction was introduced in [17], with subsequent developments for Gaussian process priors [18, 19] and L^r -metrics [20]. Building on these results, we establish analogous theory for the Tobit Gaussian process regression model with Vecchia approximation. We begin by outlining the assumptions required for our analysis:

Assumption 1 (Design). *The design points $\{x_i\}_{i=1}^n$ lie in a compact domain $\mathcal{X} \subset \mathbb{R}^d$ and are either quasi-uniform*

or drawn i.i.d. from a density bounded away from zero and infinity on $\mathcal{X}$.

Assumption 2 (True function). *The true regression function f_0 lies in a Hölder class of smoothness $\alpha > 0$ on $\mathcal{X}$ and satisfies $\|f_0\|_\infty \leq B$ for some finite B .*

Assumption 3 (Noise). *The noise $\varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma_0^2)$ has variance $\sigma_0 \in (0, \bar{\sigma}]$, and the censoring threshold $c \in \mathbb{R}$ is fixed and known.*

Assumption 4 (Prior). *The prior for f is a mean-zero Gaussian process with Matérn covariance k_θ of smoothness $\nu \geq \alpha$, and it assigns sufficient mass to Kullback–Leibler neighborhoods of f_0 .*

Assumption 5 (Vecchia approximation). *The prior distribution on $\mathbf{f} = (f(x_1), \dots, f(x_n))^\top$ is approximated by the Vecchia factorization $\tilde{p}_n(\mathbf{f}) = \prod_{i=1}^n p(f_i | f_{N(i)})$, where $N(i) \subset \{1, \dots, i-1\}$ is a conditioning set of size m_n . The approximation satisfies $\text{KL}(p(\mathbf{f}) \| \tilde{p}_n(\mathbf{f})) = o(n\varepsilon_n^2)$, with $\varepsilon_n = n^{-\alpha/(2\alpha+d)}(\log n)^t$ for some $t \geq 0$. In particular, $m_n \rightarrow \infty$ slowly with n , e.g. $m_n \asymp \log n$.*

Assumptions (1)–(4) are standard conditions under which Gaussian process priors enable minimax-optimal posterior contraction: quasi-uniform or random designs equate the empirical and population L_2 norms; Hölder regularity of f_0 ensures well-defined minimax rates; sub-Gaussian noise with a fixed censoring threshold preserves identifiability; and Matérn kernels with smoothness $\nu \geq \alpha$ place sufficient mass in Kullback–Leibler neighborhoods of f_0 . Assumption 5 is new and ensures that the Vecchia factorization approximates the full GP asymptotically. Requiring $\text{KL}(p(\mathbf{f}) \| \tilde{p}_n(\mathbf{f})) = o(n\varepsilon_n^2)$ guarantees that the approximation error is negligible relative to the information from n observations at rate ε_n . This is achieved by letting the neighborhood size m_n grow slowly with n , e.g., $m_n \asymp \log n$.

Theorem 1 (Posterior contraction). *Suppose Assumptions 1–5 hold. Let $\tilde{\Pi}(\cdot | \mathbf{y})$ denote the posterior distribution under the Tobit likelihood with censoring threshold c and the Vecchia-assisted Gaussian process prior. Then, for some sufficiently large constant $M > 0$, $\tilde{\Pi}(f : \|f - f_0\|_n \geq M\varepsilon_n | \mathbf{y}) \xrightarrow[n \rightarrow \infty]{P_{f_0}} 0$, where $\varepsilon_n = n^{-\alpha/(2\alpha+d)}(\log n)^t$ is the minimax-optimal contraction rate, up to logarithmic factors.*

Proof sketch. For space constraints we only give a sketch; the proof follows the standard Ghosal–Ghosh–van der Vaart recipe but we record the key inequalities. Write $\|f\|_n^2 = n^{-1} \sum_{i=1}^n f(x_i)^2$ and let $\varepsilon_n = n^{-\alpha/(2\alpha+d)}(\log n)^t$. Define the ball $B_n = \{f : \|f - f_0\|_n \leq M\varepsilon_n\}$ and its complement $U_n = \{f : \|f - f_0\|_n \geq M\varepsilon_n\}$ for a large $M > 0$. First, for f close to f_0 , we note that the pointwise KL/Hellinger control for the Tobit likelihood yields $\text{KL}(P_{f_0}^{(i)} \| P_f^{(i)}) \leq C_1(f(x_i) - f_0(x_i))^2$

and $h^2(P_{f_0}^{(i)}, P_f^{(i)}) \leq C_2(f(x_i) - f_0(x_i))^2$, where $C_1, C_2 > 0$ depend only on uniform bounds for f, f_0 and σ_0 . Summing over i gives $\text{KL}(P_{f_0}^{(n)} \| P_f^{(n)}) \leq C_1 n \|f - f_0\|_n^2$ and $h^2(P_{f_0}^{(n)}, P_f^{(n)}) \leq C_2 n \|f - f_0\|_n^2$. Hence, for every $f \in U_n$, we have $h^2(P_{f_0}^{(n)}, P_f^{(n)}) \geq c_0 n \varepsilon_n^2$, for some $c_0 > 0$ and M chosen large enough. Second, we construct tests ϕ_n , e.g. likelihood-ratio or Hellinger-projection tests, separating f_0 from $\{f : \|f - f_0\|_n \geq M\varepsilon_n\}$ with exponential errors: $E_{f_0}(\phi_n) \leq \exp(-C_3 n \varepsilon_n^2)$, $\sup_{f \in U_n} E_f(1 - \phi_n) \leq \exp(-C_3 n \varepsilon_n^2)$. Third, we check the prior mass condition. For the full GP prior Π , one has the usual small-ball bound $\Pi(B_n) \geq \exp(-C_4 n \varepsilon_n^2)$, and an sieve construction gives $\log N(\varepsilon_n, \mathcal{F}_n, \|\cdot\|_n) \lesssim n \varepsilon_n^2$ and $\Pi(\mathcal{F}_n^c) \leq \exp(-(C_4 + 4)n \varepsilon_n^2)$. Finally, we transfer the prior-mass bounds to the Vecchia assisted prior $\tilde{\Pi}$ using Assumption (5). Let $p(\mathbf{f})$ and $\tilde{p}_n(\mathbf{f})$ denote the exact and Vecchia finite-dimensional GP densities on $\mathbf{f} = (f(x_1), \dots, f(x_n))$. Define the log-likelihood ratio $\ell(\mathbf{f}) = \log \frac{p(\mathbf{f})}{\tilde{p}_n(\mathbf{f})}$. Then, for any measurable set A ,

$$\begin{aligned} \tilde{\Pi}(A) &= \int_A \tilde{p}_n(\mathbf{f}) d\mathbf{f} = \int_A p(\mathbf{f}) e^{-\ell(\mathbf{f})} d\mathbf{f} \\ &\geq \int_A p(\mathbf{f}) e^{-\ell(\mathbf{f})} \mathbf{1}_{\{\ell(\mathbf{f}) \leq \delta\}} d\mathbf{f} \geq e^{-\delta} \Pi(A \cap \{\ell(\mathbf{f}) \leq \delta\}), \end{aligned}$$

for any $\delta > 0$. By Markov's inequality, $\Pi(\ell(\mathbf{f}) > \delta) \leq \text{KL}(p \| \tilde{p}_n) / \delta = o(1)$ with $\delta = o(n\varepsilon_n^2)$, so $\Pi(A \cap \{\ell(\mathbf{f}) \leq \delta\}) \approx \Pi(A)$ and hence $\tilde{\Pi}(A) \geq e^{-o(n\varepsilon_n^2)} \Pi(A)$, showing that the Vecchia prior preserves the small-ball probabilities at the contraction scale up to negligible multiplicative factors. Applying this to $A = B_n$ yields $\tilde{\Pi}(B_n) \geq \exp(-C_4 n \varepsilon_n^2 - o(n\varepsilon_n^2)) = \exp(-C_5 n \varepsilon_n^2)$. Thus, the Vecchia approximation preserves the requisite small-ball mass up to negligible factors. Putting the pieces together, with high P_{f_0} -probability the posterior mass on U_n is bounded by

$$\tilde{\Pi}(U_n | \mathbf{y}) \leq \frac{\int_{U_n} \prod_{i=1}^n p_f^{(i)}(y_i) d\tilde{\Pi}(f)}{\int_{B_n} \prod_{i=1}^n p_f^{(i)}(y_i) d\tilde{\Pi}(f)} \leq \frac{\exp(-C_3 n \varepsilon_n^2) \cdot \tilde{\Pi}(U_n)}{\exp(-C_6 n \varepsilon_n^2) \cdot \tilde{\Pi}(B_n)},$$

where the exponential factor in the numerator comes from the likelihood concentration bounds and the denominator uses the lower bound on the marginal likelihood over B_n . Using $\tilde{\Pi}(U_n) \leq 1$ and the lower bound $\tilde{\Pi}(B_n) \geq \exp(-C_5 n \varepsilon_n^2)$ gives $\tilde{\Pi}(U_n | \mathbf{y}) \leq \exp(-(C_3 - C_5 - C_6)n \varepsilon_n^2 + o(n\varepsilon_n^2))$. Choosing M large makes $C_3 - C_5 - C_6 > 0$, hence the right-hand side tends to zero in P_{f_0} -probability. The same argument extends to unknown σ provided the prior on σ places sufficient mass in a neighborhood of σ_0 ; the only modification is to include σ in the KL/Hellinger bounds and in the small-ball calculations, which is routine. $\square$

4. PERFORMANCE EVALUATION

Numerical Experiment 1. We consider $f(x) = \sin(6x - 3)$ on an n -dimensional grid in $[0, 1]$, perturb the evaluations with Gaussian noise ($\sigma = 0.02$), and censor values

below -0.9 and above 0.8 , yielding approximately 35% censored observations (Figure 1a). We repeat the experi-

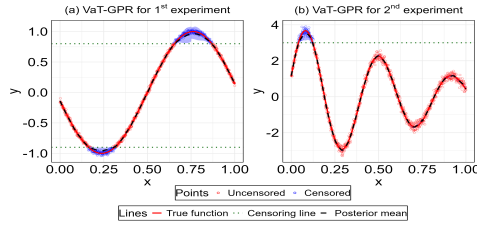


Fig. 1: Numerical experiments. *VaT-GPR* instances from each of the two experiment settings with $n = 1000$.

ment 20 times and fit both T-GPR and VaT-GPR to each dataset. In VaT-GPR, we consider $m \approx 5\log(n)$. As shown in Table 1, VaT-GPR and T-GPR achieves similar RMSE and coverage as the vanilla T-GPR for varying n . Even under $\sim 35\%$ censoring, it recovers the true function in most cases (Figure 1a). Moreover, the runtime of VaT-GPR scales nearly linearly with n , from 16 minutes for $n = 500$ to 25 minutes for $n = 1000$, whereas T-GPR becomes computationally prohibitive.

n	Method	Experiment 1		Experiment 2	
		RMSE (% coverage.)	Time (hrs)	RMSE (% coverage.)	Time (hrs)
500	T-GPR	.027 (88)	1.36	.054 (95)	1.37
	VaT-GPR	.021 (89)	0.27	.059 (96)	0.29
600	T-GPR	.025 (88)	1.52	.058 (93)	1.55
	VaT-GPR	.021 (90)	0.33	.061 (94)	0.34
800	T-GPR	.028 (90)	4.77	.056 (95)	4.11
	VaT-GPR	.023 (90)	0.42	.064 (95)	0.47
1000	T-GPR	.028 (90)	10.61	.053 (95)	10.97
	VaT-GPR	.025 (90)	0.46	.056 (95)	0.46

Table 1: Numerical experiments. *RMSE and Computation time for T-GPR and VaT-GPR for varying n .*

Numerical Experiment 2. We consider $f(x) = (x - 2)^2 \sin(3(5x - 2))$ on an n -dimensional grid in $[0, 1]$, perturb the evaluations with Gaussian noise ($\sigma = 0.14$), and right-censor values above 3, yielding approximately 9% censored observations (Figure 1b). As shown in Table 1, VaT-GPR and T-GPR fare comparably with respect to RMSE and coverage, while VaT-GPR yields massive computation gains. Further, the 95% credible intervals yielded by VaT-GPR mostly contain the true function (Figure 1b).

Analysis of NO_x concentration data. We analyze nitrogen oxides (NO_x) concentration data from the Marylebone super-site in London, spanning January 1998 to June 2005, collected by the London Air Quality Archive and made available through the `openair` package in R [21]. NO_x comprises nitric oxide (NO) and nitrogen dioxide (NO_2), two highly toxic gases produced as byproducts of combustion processes, e.g., in power plants, with adverse effects on environmental and biological systems, including ozone layer depletion and acid rain.

We first compute the daily averages of the $\log_{10}$ -transformed NO_x concentrations [22], which stabilizes

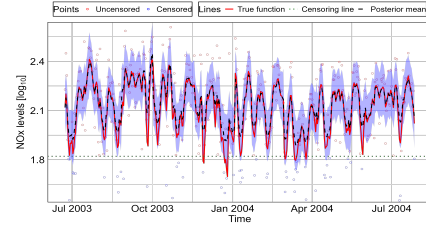


Fig. 2: NO_x concentration data. *A segment of the observed $\log_{10}$ -transformed NO_x concentrations from July 2003 to July 2004, along with the posterior mean and 95% credible interval.*

variance and mitigates the influence of extreme values. We then fit a VGPR model [23] to the resulting 2400 observations and treat the posterior mean (solid red line in Figure 2) as the true underlying function. To emulate left-censored observations, we identify the 10%-tile of the data and artificially censor all values below this threshold, thereby mimicking the effect of detection limits commonly encountered in environmental monitoring.

	$m = 4$	$m = 10$	$m = 25$	$m = 30$	$m = 35$	$m = 50$
RMSE	0.13	0.07	0.04	0.04	0.04	0.04
Time (hr)	1.2	1.4	2.5	2.8	3.2	4.2

Table 2: NO_x concentration data. *RMSE and computation time across different values of m .*

Figure 2 displays the posterior mean and 95% credible interval obtained from fitting VaT-GPR with $m = 30$. The credible interval captures the true underlying function across most regions. Moreover, for VaT-GPR, 4000 MCMC iterations run within 2.8 hours on a personal laptop, whereas the vanilla implementation of T-GPR cannot be executed due to excessive memory requirements. This demonstrates that VaT-GPR is highly scalable and practical in scenarios where T-GPR is computationally infeasible. Further, Table 2 indicates that the posterior estimates stabilize for moderate values of conditioning set size m , as anticipated.

CONCLUSION

We propose VaT-GPR, a scalable and computationally efficient extension of T-GPR for censored functional responses. By leveraging parallel architectures and sparse Cholesky factorization of the GP covariance, the method achieves linear memory and computational complexity. Code is available on GitHub. Future work includes extensions to multi-output censored GPs, variable selection, and spatially censored data analysis.

Acknowledgement. Anirban Chakraborty’s research has been supported by Institutional Research Grant #IRG-24-1290553-23-IRG, from the American Cancer Society.

5. REFERENCES

- [1] B. Allik, C. Miller, M. J. Piovoso, and R. Zurakowski, "The Tobit Kalman filter: An estimator for censored measurements," *IEEE Transactions on Control Systems Technology*, vol. 24, no. 1, pp. 365–371, 2015.
- [2] D. Gammelli, K. P. Rolsted, D. Pacino, and F. Rodrigues, "Generalized multi-output gaussian process censored regression," *Pattern Recognition*, vol. 129, p. 108751, 2022.
- [3] S. Chib, "Bayes inference in the Tobit censored regression model," *Journal of Econometrics*, vol. 51, no. 1-2, pp. 79–99, 1992.
- [4] C. G. Kaufman, M. J. Schervish, and D. W. Nychka, "Covariance tapering for likelihood-based estimation in large spatial data sets," *Journal of the American Statistical Association*, vol. 103, no. 484, pp. 1545–1555, 2008.
- [5] A. Datta, S. Banerjee, A. O. Finley, and A. E. Gelfand, "Hierarchical nearest-neighbor Gaussian process models for large geostatistical datasets," *Journal of the American Statistical Association*, vol. 111, no. 514, pp. 800–812, 2016.
- [6] M. Katzfuss and J. Guinness, "A general framework for Vecchia approximations of Gaussian processes," *Statistical Science*, vol. 36, no. 1, pp. 124–141, 2021.
- [7] L. Wu, G. Pleiss, and J. P. Cunningham, "Variational nearest neighbor Gaussian process," in *International Conference on Machine Learning*, pp. 24114–24130, PMLR, 2022.
- [8] T. Gyger, R. Furrer, and F. Sigrüst, "Vecchia-inducing-points full-scale approximations for gaussian processes," 2025.
- [9] A. V. Vecchia, "Estimation and Model Identification for Continuous Spatial Processes," *Journal of the Royal Statistical Society. Series B (Methodological)*, vol. 50, no. 2, pp. 297–312, 1988.
- [10] M. Katzfuss, J. Guinness, W. Gong, and D. Zilber, "Vecchia Approximations of Gaussian-Process Predictions," *JABES*, vol. 25, pp. 383–414, 2020.
- [11] A. Chakraborty and A. Chakraborty, "Scalable Model-Based Gaussian Process Clustering," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5730–5734, 2024.
- [12] M. Basson, T. M. Louw, and T. R. Smith, "Variational tobit gaussian process regression," *Statistics and Computing*, vol. 33, no. 3, p. 64, 2023.
- [13] T. M. Davies, S. Banerjee, A. P. Martin, and R. E. Turnbull, "A nearest-neighbour Gaussian process spatial factor model for censored, multi-depth geochemical data," *Journal of the Royal Statistical Society Series C: Applied Statistics*, vol. 71, no. 4, pp. 1014–1043, 2022.
- [14] J. Guinness, "Permutation and Grouping Methods for Sharpening Gaussian Process Approximations," *Technometrics*, vol. 60, pp. 415–429, 10 2018.
- [15] M. L. Stein, "2010 Rietz lecture: When does the screening effect hold?," *Annals of Statistics*, vol. 39, no. 6, pp. 2795–2819, 2011.
- [16] F. Schäfer, M. Katzfuss, and H. Owhadi, "Sparse cholesky factorization by kullback–leibler minimization," *SIAM Journal on Scientific Computing*, vol. 43, no. 3, pp. A1606–A1632, 2021.
- [17] S. Ghosal, J. K. Ghosh, and A. van der Vaart, "Convergence rates of posterior distributions," *Annals of Statistics*, vol. 28, no. 2, pp. 500–531, 2000.
- [18] A. van der Vaart and J. H. van Zanten, "Rates of contraction of posterior distributions based on Gaussian process priors," *Annals of Statistics*, vol. 36, no. 3, pp. 1435–1463, 2008.
- [19] A. van der Vaart and J. H. van Zanten, "Adaptive Bayesian estimation using a Gaussian random field with inverse Gamma bandwidth," *Annals of Statistics*, vol. 37, no. 5B, pp. 2655–2675, 2009.
- [20] E. Giné and R. Nickl, "Rates of contraction for posterior distributions in l^r -metrics, $1 \leq r \leq \infty$," *Annals of Statistics*, vol. 40, no. 2, pp. 907–941, 2012.
- [21] D. C. Carslaw and K. Ropkins, "openair — An R package for air quality data analysis," *Environmental Modelling & Software*, vol. 27–28, no. 0, pp. 52–61, 2012.
- [22] C. Leigh, S. Kandanaarachchi, J. M. McGree, R. J. Hyndman, O. Alsibai, K. Mengersen, and E. E. Peterson, "Predicting sediment and nutrient concentrations from high-frequency water-quality data," *PLoS One*, vol. 14, no. 8, p. e0215503, 2019.
- [23] J. Cao, J. Guinness, M. G. Genton, and M. Katzfuss, "Scalable Gaussian-process regression and variable selection using Vecchia approximations," *Journal of machine learning research*, vol. 23, no. 348, pp. 1–30, 2022.